Charting modeled weather ranges – a microproject
Nov 12, 2025

Background

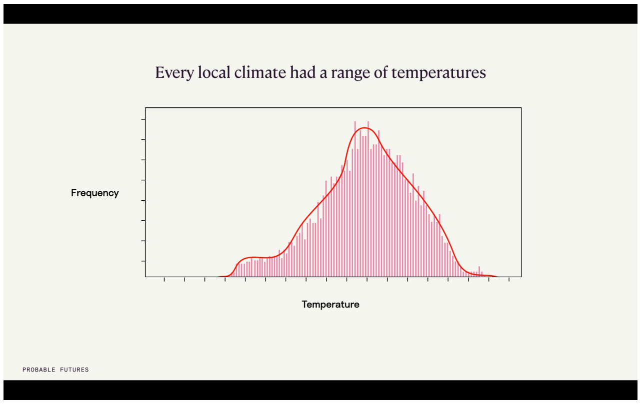
In presentations, we use histogram charts showing how temperature and precipitation are shifting. These charts are intuitive. They can convey the past, present and future likelihood of a range of temperatures or precipitation succinctly, along with the shifting likelihood at different warming scenarios. They show how those shifts often mean fewer days with moderate temperatures or precipitation and more with extremes – visually, as widening and flattening curves. And they also can demonstrate how climate models work by showing individual days from simulated years.

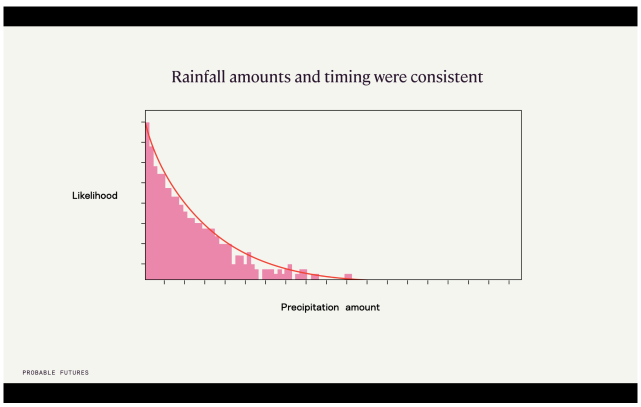
The idea behind this project is: What if we could easily see charts like this for any place? These charts could be used in presentations as slides, or even used in real time.

This is the kind of information that would be suitable in the original “climate factbook” idea. The current iteration of the factbook uses a lot of maps, plus cumulative distribution charts that are not as intuitive as histograms or charts of raw climate data. With charts like the ones below, we could “de-mapify” the factbook, focusing it on climate facts as was originally envisioned.

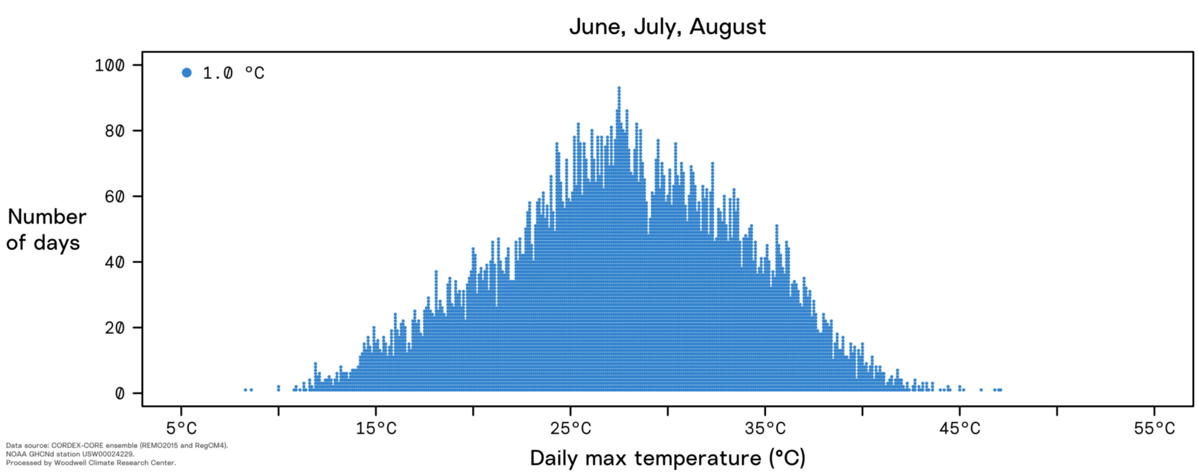
Example charts

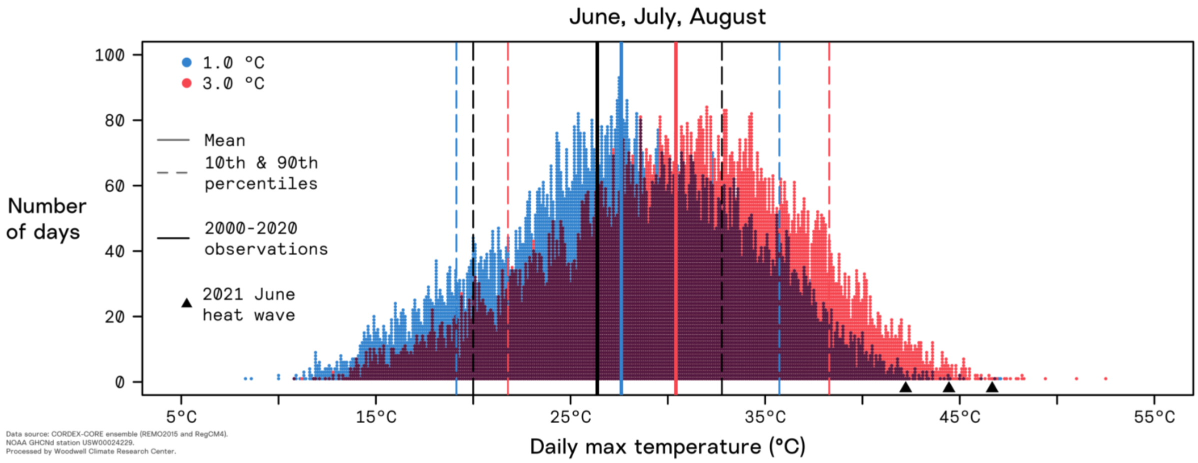
In presentations, we use binned histogram charts like this:





Other charts we have used look like this:





Scope and effort

Because modeled climate data is so large, the PF team has been advised in the past that charts like this on-the-fly are impossible. What changed? Moustafa and Peter understand the data better now, and understand that there are ways to structure and organize it that make charting manageable. Moustafa also has more experience working with big data than he did in the past. And, AI chat tools make estimating the scope and feasibility of a project like this much easier.

The scope of this project would involve getting raw climate model datasets from Carlos for tmax, tmin, precip, tmax wet-bulb, tmin wet-bulb, and maybe heat waves. These are the raw outputs from the three REMO and three RegCM models that are used in the ensemble used on Probable Futures climate maps. Then, Moustafa would put that data in a database and build a front-end where the data from any location could be rendered as a chart like one of the charts above (e.g. in the factbook). We expect this would take one to three days of Carlos's time and would fit easily within Mosutafa's existing responsibilities maintaining our systems and building the AI prototype due to his familiarity with the data and creating charts such as those in the factbook.

Scope of the data

How much data are we talking about? For each variable (precip, tmax wet-bulb, tmin wet-bulb, and maybe heat waves), here are the numbers.

In each grid cell, using three models from REMO and three from RegCM:
365 days x 21 years x 6 models = 45,990 modeled days for each warming scenario
365 days x 30 years x 6 models = 65,700 modeled days for the baseline

There are about 500,000 grid cells in our maps, once the oceans are removed.
45,990 x 500,000 = 22,995,000,000 modeled days globally (per non-baseline warming scenario)
65,700 x 500,000 = 32,850,000,000 modeled days globally (for the baseline)

There are five non-baseline warming scenarios
5 x 22995000000 = 114,975,000,000

Putting it all together:
114,975,000,000 + 32,850,000,000 = 1,478,250,000,00

Storing the data

Moustafa estimates this amount of data is about 7TB.

He sketched out a database structure with the following columns: long, lat, warming_scenario, tmax, tmin, precip, tmax_wb, tmin_wb. Structuring it this way gives us a database with 18,000,000 rows. We could potentially partition the database into six virtual tables with 3 million rows each, for performance.

In our database on AWS, this would cost about $600/mo in storage costs. The charts would render faster (likely less than one second on a modern device) with this option.

With file storage like S3 parquet (file format) and athena (server running multiple workers that can run multiple queries at a time), it would likely be a fraction of that cost, but potentially render more slowly, such as 1-2 seconds.

Rendering in the browser

For any given location, we would be rendering either one or two scenarios in the browser. That would mean rendering 45,990 dots (representing modeled days for each warming scenario) or 65,700 dots (representing modeled days for the baseline).

Not all rendering options are the same. ChatGPT provides this advice:

If you use WebGL (e.g., scattergl in Plotly, or Deck.gl's scatter/heatmap layers), 65,700 points is well within feasible range. Rendering and interactivity should both be smooth. To render multiple warming scenarios at a time, we could consider adding an option to switch between "raw" and "smoothed/histogram" modes.

Step	Action	Time estimate
Query Factbook API	Retrieve 65,700 daily points	0.2–0.5 s
Network transfer	0.5–1 MB	0.1–0.2 s
JSON parse & render to WebGL	65k floats → GPU	0.3–0.7 s
Total perceived load time		**≈ 0.6–1.2 seconds**

Storage and pre-processing test plan

We will begin testing with the North America domain. Our goal will be to determine the most efficient way to store and query the data to reduce cost and latency when a user searches for climate information about a place.

Moustafa came up with this plan to test. If this plan works well, Carlos can provide the data in the specific format of the partitioned parquet file, specified in this diagram. That way, "glue" steps in the diagram can be omitted, and the data will be smaller – in parquet format, about half the size of CSV format.

Normalizing 0.5 vs other warming scenarios

Since the 0.5°C baseline scenario has ~30 observed simulated years and other scenarios have ~21 such years, if we render charts where the height of the bars is equal to the exact number of observed days, the 0.5°C chart will have erroneously longer bars.

To solve this and enable easy comparison between 0.5°C and other scenarios, we will normalize the height of the bars in the 0.5°C charts based on the ratio of total observed days between the two scenarios.

For example, for New York City, consider a chart that compares 0.5°C and 2°C. It has 65424 observed days in the 0.5°C scenario and 45798 such days in the 2°C scenario. To normalize 0.5°, we find the quotient of the two numbers and use the result to adjust the height of the 0.5°C bars accordingly.

45798 / 65424 = 0.7000183419

Thus, the bars of the 0.5°C scenario for New York City will be about 0.7 (exactly 0.7000183419) of the height that they would be if they were exactly the height of their observed days count.